

Week-8

→ for the following sequence of pages
using fifo, lru, mru, optimal with physical
memory size ~~with~~ 3 frames

Sequence :- 3 2 1 3 4 1 6 2 4 3 4 2 1 4 5 2 1 3 4

i) FIFO :-

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
			1	1	1	1	1	2	2	2	2	2	1	1	1	1	1	3	3
		2	2	2	2	2	6	6	6	6	4	4	4	4	4	2	2	2	2
	3	3	3	3	4	4	4	4	4	3	3	3	3	3	5	5	5	5	4
Page fault X → miss ✓ → hit	X	X	X	✓	X	✓	X	X	✓	X	X	✓	X	✓	X	X	✓	X	X

$$\text{Hit Ratio} = \frac{6}{19}$$

ii) +

$$\begin{aligned} \text{Miss Ratio} &= 1 - \frac{6}{19} \\ &= \frac{19-6}{19} = \frac{13}{19} \end{aligned}$$

H. Z. P. L. e. n. e
ii) efficiency
in) Bels

ii) LRU :-

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
			1	1	4	4	1	1	4	4	4	4	1	1	1	2	2	3	3
		2	2	2	2	4	4	4	2	2	2	2	2	2	5	5	5	5	4
	3	3	3	3	3	3	6	6	6	3	3	3	3	4	4	4	1	1	1
	3	3	3	3	3	3	X	X	X	X	✓	✓	X	X	X	X	X	X	X
X	X	X	✓	X	✓	X	X	X	X	X	✓	✓	X	X	X	X	X	X	X

$$\text{Hit Ratio} = \frac{4}{19}$$

$$\text{Miss Ratio} = 1 - \frac{4}{19} = \frac{19-4}{19} = \frac{15}{19}$$

FIFO has 13/19 Page faults & 6/19 Page hits, LRU has 14/19 faults and 5/19 faults, MRU has 12/19 faults and 7/19 hits and optimal has 10/19 faults and 9/19 hits

So based on page fault (lower is better to choose):

optimal < MRU < FIFO < LRU

- 2) Apply FIFO page Replacement algorithm for the sequence pages and check whether Belady's anomaly exists or not.

For the Belady's take the physical memory as 3 frame's and 4 frame's

Sequence: 1 2 3 4 1 2 5 1 2 3 4 5

→ FIFO with 3

	1	2	3
			3
		2	2
	1	1	1
Page fault	X	X	X

X → miss
✓ → Hit

→ FIFO with

	1	2	3
			3
	1	2	2
	1	1	1
Page fault	X	X	X

In FIFO while 4 number of which

14/19 faults

→ FIFO with 3 frames.

	1	2	3	4	1	2	5	1	2	3	4	5
			3	3	3	2	2	2	2	2	4	4
		2	2	2	1	1	1	1	1	3	3	3
	1	1	1	4	4	4	5	5	5	5	5	5
Page fault x → miss ✓ → Hit	x	x	x	x	x	x	x	✓	✓	x	x	✓

$$\text{Page hits} = 3/12$$

$$\begin{aligned} \text{Page faults} &= 1 - \frac{3}{12} \\ &= 9/12 \end{aligned}$$

→ FIFO with 4 Frames:

	1	2	3	4	1	2	5	1	2	3	4	5
				4	4	4	4	4	4	3	3	3
			3	3	3	3	3	2	2	2	2	2
	1	2	2	2	2	2	1	1	1	1	5	5
	1	1	1	1	1	5	5	5	5	4	4	4
	x	x	x	x	✓	✓	x	x	x	x	x	x

$$\text{Page hits} = 2/12$$

$$\text{Page faults} = 1 - \frac{2}{12} = 10/12$$

In FIFO Page replacement, 3 frames produce 9 page faults, while 4 frames produce 10 Page faults. Thus, increasing the number of frames increases the number of page faults, which is called Belady's Anomaly. So Belady's exists.

for the
other before

iii) MRU:- most Recently used:-

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
			1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6
			2	2	2	2	2	2	2	2	2	2	1	1	1	1	1	1
			3	3	3	3	4	4	4	4	3	4	4	4	5	2	2	2
x	x	x	x	✓	x	✓	x	✓	✓	x	x	✓	x	✓	x	x	✓	x

Page faults = 12

Page hits = $\frac{7}{19}$

$$\begin{aligned} \text{Page faults} &= 1 - \frac{7}{19} \\ &= \frac{19-7}{19} \\ &= \frac{12}{19} \end{aligned}$$

iv) optimal:-

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
			1	1	1	1	6	6	6	3	3	3	1	1	1	1	1	1
			2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
			3	3	3	3	4	4	4	4	4	4	4	5	5	5	5	3
x	x	x	x	✓	x	✓	x	✓	✓	x	✓	✓	x	✓	x	✓	✓	x

Page hits = 9/19

$$\text{faults} = 1 - \frac{9}{19} = \frac{10}{19}$$